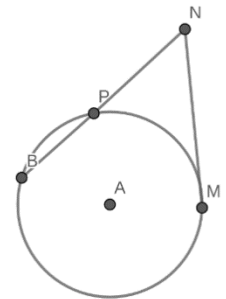
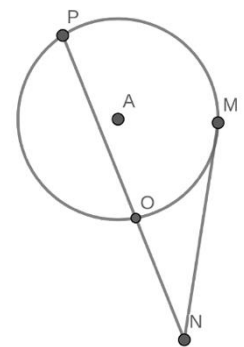


1. Circle A is shown with points M , B and P on the circle. Secant $\overline{BN}$ and tangent $\overline{MN}$ intersect at point N outside the circle. $PB = 4$ and $PN = 9$. What is the length of $\overline{MN}$? Round to the nearest thousandth.

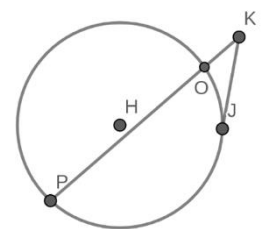


2. Circle A is shown with points M , O and P on the circle. Secant $\overline{PN}$ and tangent $\overline{MN}$ intersect at point N outside the circle. $PO = (2x + 3)$, $ON = 11$ and $MN = 22$. Find the value of x .



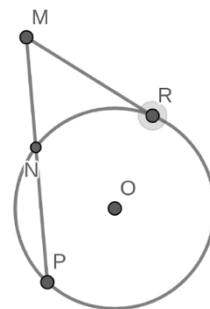
3. What is the length of $\overline{PN}$?

4. Circle H is shown with points J , O and P on the circle. Secant $\overline{KP}$ and tangent $\overline{KJ}$ intersect at point K outside the circle. $PO = 120$, $OK = 8$ and $KJ = x$. Find the value of x .



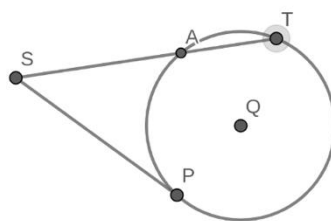
5. Using the same image as question 4, circle H is shown with points J , O , and P on the circle. Secant $\overline{KP}$ and tangent $\overline{KJ}$ intersect at point K outside the circle. $PO = 12 + x$, $OK = 10$ and $KJ = 18$. Find the length of $\overline{KP}$.

6. Circle O is shown with points N , R and P on the circle. Secant $\overline{MP}$ and tangent $\overline{MR}$ intersect at point M outside the circle. $PN = 16$, $MN = 10$ and $MR = 2x$. What is the value of x ? Round to the nearest thousandth.

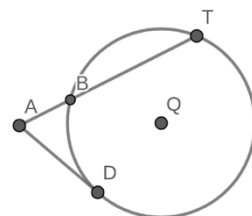


7. Using the same image as question 6, Circle O is shown with points N , R and P on the circle. Secant $\overline{MP}$ and tangent $\overline{MR}$ intersect at point M outside the circle. $PN = x + 17$, $MN = 12$ and $MR = 24$. What is the value of x ?

8. Circle Q is shown with points A , T and P on the circle. Secant $\overline{ST}$ and tangent $\overline{SP}$ intersect at point S outside the circle. $PS = 6x$, $AT = 10$ and $AS = 25$. What is the value of x ? Round to the nearest thousandth.



9. Daniella and Taylor are working on finding the length of $\overline{AD}$. Circle Q is shown with points B , T and D on the circle. Secant $\overline{AT}$ and tangent $\overline{AD}$ intersect at point A outside the circle. $BT = 15$, $AB = 12$ and $AD = 2x$. Their partial work solving for x is shown below. Whose work do you agree with?



Daniella's work	Taylor's work
$(2x)^2 = 15(15 + 12)$ $4x^2 = 15(27)$ $4x^2 = 405$ $x^2 = 101.25$	$(2x)^2 = 12(15 + 12)$ $4x^2 = 12(27)$ $4x^2 = 324$ $x^2 = 81$

10. Using the image in Question 9, what is the length of $\overline{AD}$?